

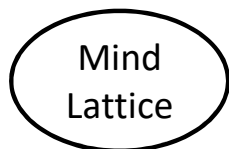
# SMART INDIA HACKATHON 2026



## TITLE PAGE

- **Problem Statement ID –**
- **Problem Statement Title-**
- **Theme-**
- **PS Category- Software/Hardware**
- **Team ID-**
- **Team Name (Registered on portal)**





# IDEA TITLE



## Proposed Solution — MediKiosk.

**MediKiosk** is an AI-powered, multilingual patient history-taking kiosk that transforms patient inputs, medical records, and vitals into a structured, doctor-ready clinical case.

### 1. Nurse Screening

Nurse records **vitals (BP, temperature)** → system creates a **unique Visit ID**.

### 2. AI Patient Interview

Patient selects **language + consultation type (General Medicine/Ayurveda)** → AI conducts an adaptive interview through **voice, text, or touch**.

### 3. Multimodal Medical History

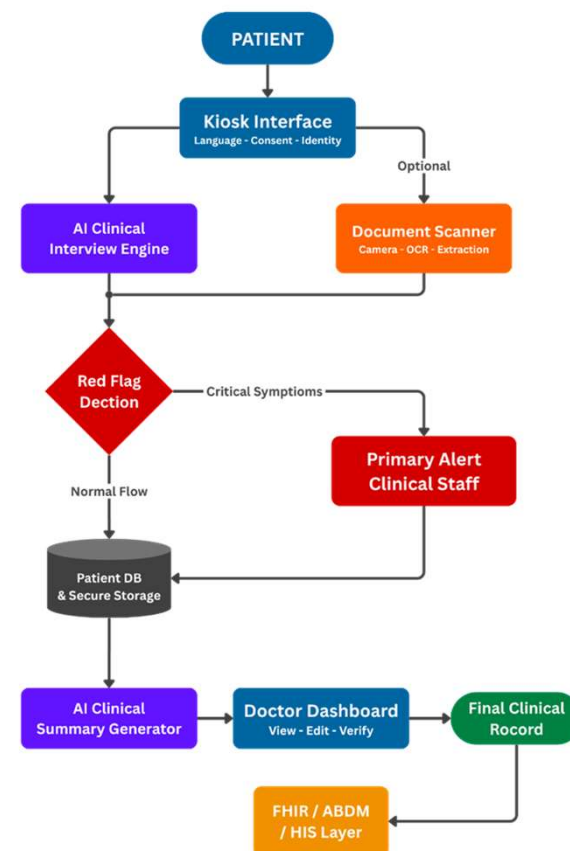
Collects **patient responses + prescriptions/reports/discharge summaries** → OCR and AI extract and structure relevant medical information into a **medical timeline**.

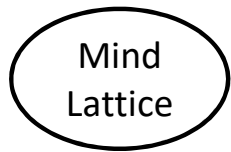
### 4. Smart Prioritization

Combines **symptoms, history, and vitals** → rule-based red-flag detection categorizes cases as **Normal, High Priority, or Critical** to assist staff.

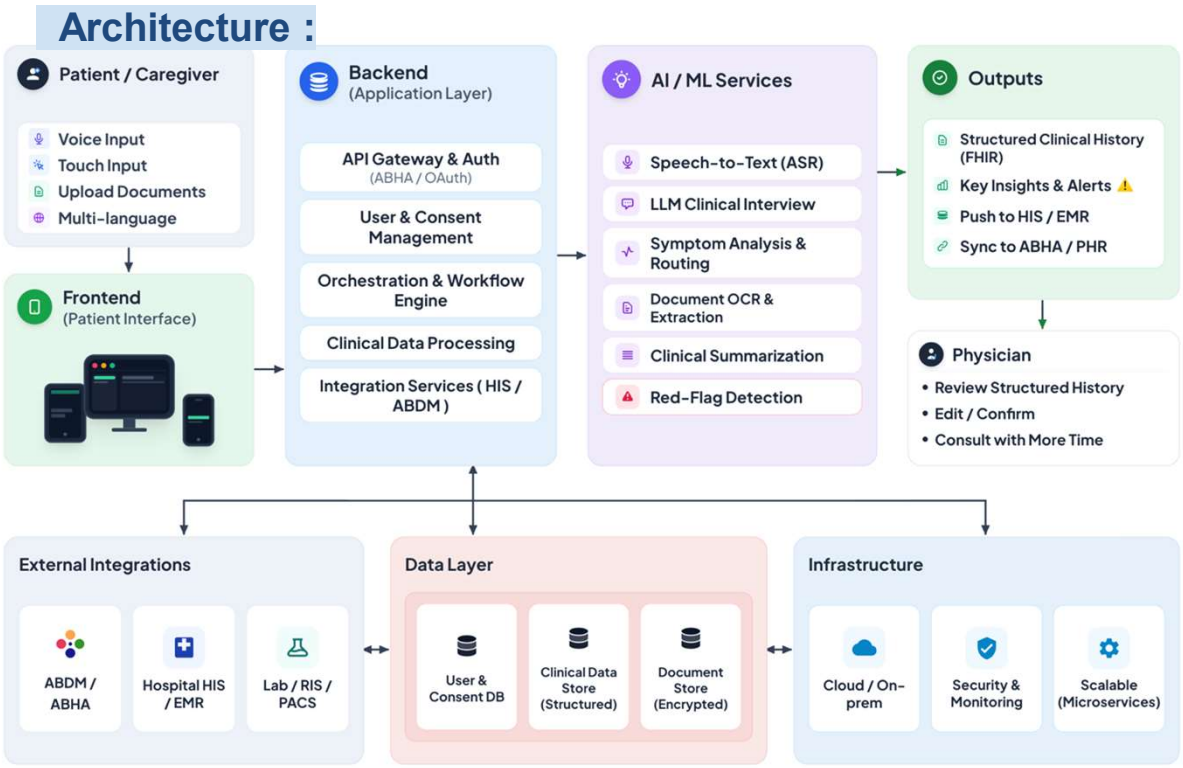
### 5. Doctor-Ready Case

Doctor receives a concise dashboard with **complaints, history, medications, allergies, investigations, vitals, timeline, red flags, and source documents** → **reviews, verifies, edits, and makes the final clinical decision**.



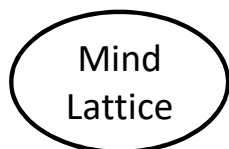


# TECHNICAL APPROACH



**Technologies to be Used :**

| FRONTEND                                                                                                                               | BACKEND                                                                                                                   | AI / ML                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>React.js</li><li>Next.js</li><li>TypeScript</li><li>Tailwind CSS</li><li>Framer Motion</li></ul> | <ul style="list-style-type: none"><li>Node.js</li><li>Next.js API</li><li>Python</li><li>FastAPI</li><li>Render</li></ul> | <ul style="list-style-type: none"><li>Gemini Gemini API</li><li>OCR / Doc AI</li><li>Speech-to-Text</li><li>Text-to-Speech</li><li>Translation AI</li></ul> |
| DATABASE                                                                                                                               | CLOUD / INFRA                                                                                                             | HEALTHCARE                                                                                                                                                  |
| <ul style="list-style-type: none"><li>PostgreSQL</li><li>Supabase</li><li>Supabase Storage</li></ul>                                   | <ul style="list-style-type: none"><li>Vercel</li><li>GitHub</li><li>Docker</li><li>GitHub Actions</li></ul>               | <ul style="list-style-type: none"><li>HL7 FHIR</li><li>Google Doc AI</li><li>Hospital EMR API</li><li>REST APIs</li></ul>                                   |



# FEASIBILITY AND VIABILITY



## 1. FEASIBILITY

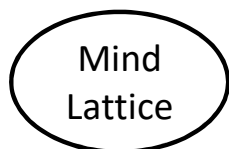
- **Technical:** React + Vite, Gemini, OCR, Firebase & Cloud Run provide a scalable foundation.
- **Economic:** Software-first approach works with existing PCs/tablets/kiosks, minimizing hardware and infrastructure costs.
- **Operational:** Fits the workflow **Patient** → **Nurse** → **MediKiosk** → **Doctor** with voice, text and touch interfaces.
- **Healthcare Integration:** FHIR R4 + ABDM-ready APIs enable future integration with hospital systems.
- **Scalable:** Can be deployed across **OPDs, government hospitals, medical colleges and AYUSH hospitals.**

## 2. CHALLENGES & RISKS

- **AI errors** or incorrect symptom interpretation
- OCR errors in **poor-quality medical documents**
- Language, accent and speech-recognition issues
- Patient data **privacy and security**
- Resistance to adopting a new workflow

## 3. STRATEGIES TO OVERCOME

- **AI Accuracy** → Rule-based validation + confidence checks + doctor verification
- **OCR Errors** → Document quality checks + manual correction
- **Language Barriers** → Multilingual AI + voice/text/touch fallback
- **Data Security** → Encryption + role-based access + audit logs
- **Adoption** → Simple UI + minimal training + nurse-assisted onboarding



# IMPACT AND BENEFITS



## Potential Impact on Target Audience

### Patients:

- **Shorter waiting & consultation time** through faster history collection
- **Multilingual and accessible** interaction using voice, text and touch
- Better continuity of care through a **structured medical history**

### Nurses & Hospital Staff:

- Reduces **repetitive history-taking and paperwork**
- Helps identify **high-priority cases earlier**
- Frees staff to focus on **patient care**

### Doctors:

- Receives a **concise, structured clinical case** before consultation
- Easier access to **past records, medications, investigations and vitals**
- More time for **clinical assessment and decision-making**

## Broader Benefits

### Social:

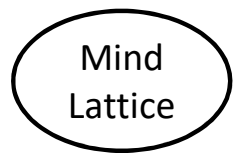
- Improves access for **regional-language users**
- Supports more consistent **patient screening**
- Improves **continuity of care**

### Economic:

- Reduces **manual documentation workload**
- Makes efficient use of **clinical staff time**
- Scalable across hospitals without **major specialized hardware**

### Environmental:

- Reduces **paper-based records**
- Less **printing, storage and physical record handling**
- Enables more **digital-first healthcare workflows**



# RESEARCH AND REFERENCES



- Details / Links of the reference and research work

# IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. ( Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

**Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.**